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DBMS Lab Assignment - 4

Group by, order by and having clause implementation for nested queries and SQL Views

Aggregate functions are performed on the columns values. But if we want to divide values stored in a single column in groups and then perform certain actions, then we have to use Group By clause.

GROUP BY is used to **group** values from a column, and, if you wish, performs calculations on that column. You can use COUNT, SUM, AVG, etc., functions on the grouped column.

Syntax

The syntax for the GROUP BY clause in MySQL is:

```
SELECT expression1, expression2, ... expression_n,  
       aggregate_function (expression)  
FROM tables  
[WHERE conditions]  
GROUP BY expression1, expression2, ... expression_n;
```

Parameters or Arguments

expression1, expression2, ... expression_n

The expressions that are not encapsulated within an aggregate function and must be included in the GROUP BY clause.

aggregate_function

A function such as SUM, COUNT, MIN, MAX, or AVG functions.

tables

The tables that you wish to retrieve records from. There must be at least one table listed in the FROM clause.

WHERE conditions

Optional. The conditions that must be met for the records to be selected.

This GROUP BY example uses the MAX function to return the name of each department and the maximum salary in the department.

```
SELECT department, MAX(salary) AS "Highest salary"  
FROM employees  
GROUP BY department;
```

Having Clause: - If we want to apply certain condition on which values should get included in the group, we have to use the "Having Clause".

Syntax

The syntax for the HAVING Clause in MySQL is:

```
SELECT expression1, expression2, ... expression_n,  
       aggregate_function (expression)  
FROM tables  
[WHERE conditions]  
GROUP BY expression1, expression2, ... expression_n  
HAVING condition;
```

Parameters or Arguments

aggregate_function

A function such as SUM, COUNT, MIN, MAX, or AVG functions.

expression1, expression2, ... expression_n

The expressions that are not encapsulated within an aggregate function and must be included in the GROUP BY clause.

WHERE conditions

Optional. These are the conditions for the records to be selected.

HAVING condition

This is a further condition applied only to the aggregated results to restrict the groups of returned rows. Only those groups whose condition evaluates to TRUE will be included in the result set.

Example - Using SUM function

Let's look at a MySQL HAVING clause example that uses the [SUM function](#).

You could also use the [SUM function](#) to return the name of the product and the total quantity (for that product). The MySQL HAVING clause will filter the results so that only products with a total quantity greater than 10 will be returned.

```
SELECT product, SUM(quantity) AS "Total quantity"  
FROM order_details  
GROUP BY product  
HAVING SUM(quantity) > 10;
```

Order By:-

The MySQL ORDER BY clause is used to sort the records in your result set.

Syntax

The syntax for the ORDER BY clause in MySQL is:

```
SELECT expressions
FROM tables
[WHERE conditions]
ORDER BY expression [ ASC | DESC ];
```

Example:

```
SELECT last_name, first_name, city
FROM contacts
WHERE last_name = 'Johnson'
ORDER BY city DESC;
```

This MySQL ORDER BY example would return all records sorted by the *city* field in descending order.

We can also set a limit on the number of values which we want to be returned by the execution of order by query.

For example:

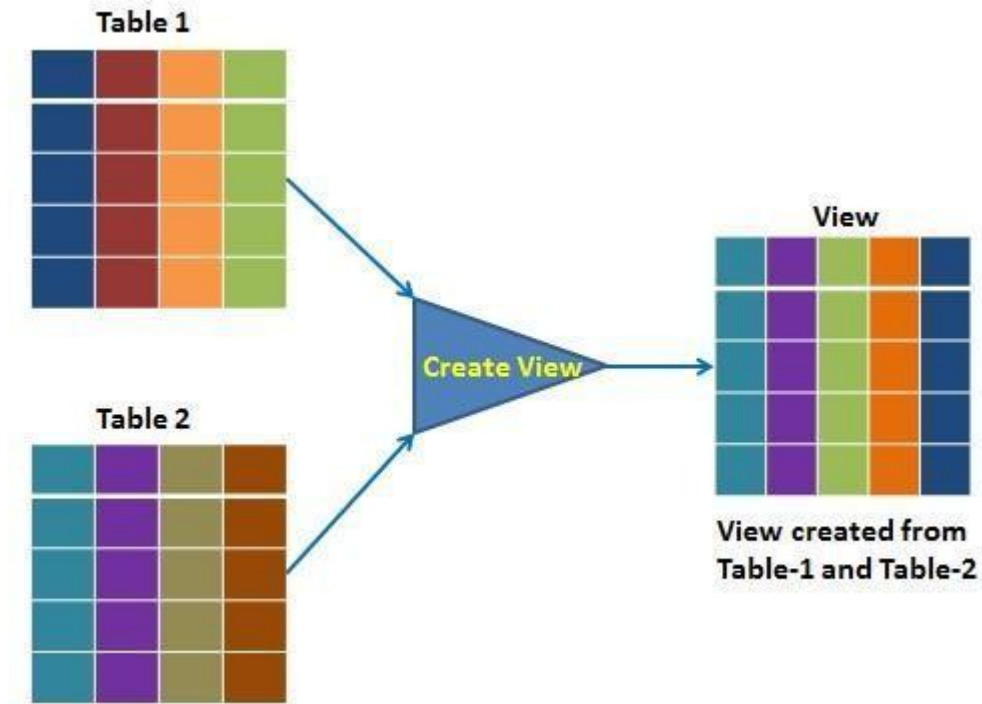
```
SELECT contact_id, last_name, first_name
FROM contacts
WHERE website = 'TechOnTheNet.com'
ORDER BY contact_id DESC
LIMIT 5;
```

This MySQL SELECT LIMIT example would select the first 5 records from the *contacts* table where the *website* is 'TechOnTheNet.com'. Note that the results are sorted by *contact_id* in descending order so this means that the 5 largest *contact_id* values will be returned by the SELECT LIMIT statement.

SQL Views

In SQL, a view is a virtual table based on the result-set of an SQL statement.

A view contains rows and columns, just like a real table. The fields in a view are fields from one or more real tables in the database.



You can add SQL functions, WHERE, and JOIN statements to a view and present the data as if the data were coming from one single table.

CREATE VIEW Syntax

```
CREATE VIEW view_name AS  
SELECT column1, column2, ...  
FROM table_name  
WHERE condition;
```

Note: A view always shows up-to-date data! The database engine recreates the data, using the view's SQL statement, every time a user queries a view.

Types of Views

- **Simple View:** A view based on only a single table, which doesn't contain GROUP BY clause and any functions.
- **Complex View:** A view based on multiple tables, which contain GROUP BY clause and functions.
- **Inline View:** A view based on a subquery in FROM Clause, that subquery creates a temporary table and simplifies the complex query.
- **Materialized View:** A view that stores the definition as well as data. It creates replicas of data by storing it physically.

Assignment 4 Lab Questions

Perform following queries using Group by, having and Order By

1. Give library wise book details.

```
mysql> SELECT
->     l.lname AS library_name,
->     COUNT(b.Bid) AS total_books
-> FROM ilibrary l
-> JOIN books b ON l.Lid = b.Lid
-> GROUP BY l.lname
-> ORDER BY l.lname ASC;
```

library_name	total_books
SCHC Library	1
SIBM Library	3
SICSR Library	1
SIOM Library	1
SIT Library	7
SLS Library	2
SSBS Library	2

7 rows in set (0.01 sec)

2. Give bookwise total copies which are available.

```
mysql> SELECT
->     b.Bname,
->     SUM(n.no_of_copies) AS total_copies
-> FROM books b
-> JOIN noofcopies n ON b.Bid = n.bid
-> GROUP BY b.Bname
-> ORDER BY total_copies DESC;
```

Bname	total_copies
Software Engineering	15
Data Structures Using C	12
Business Analytics	9
Database Management Systems	8
Operating System Concepts	7
Cloud Computing Essentials	6
Constitutional Law	5
Advanced Artificial Intelligence	4
Introduction to Algorithms	3
Computer Networks	2

10 rows in set (0.00 sec)

3. Which library has total copies more than 100?

```
mysql> SELECT
->     l.lname AS library_name,
->     SUM(n.no_of_copies) AS total_copies
-> FROM ilibrary l
-> JOIN noofcopies n ON l.Lid = n.blid
-> GROUP BY l.lname
-> HAVING SUM(n.no_of_copies) > 100
-> ORDER BY total_copies DESC;
+-----+-----+
| library_name | total_copies |
+-----+-----+
| SIT Library  |          133 |
+-----+-----+
1 row in set (0.00 sec)
```

4. Give institute wise department details.

```
mysql> SELECT
->     Iname AS institute_name,
->     COUNT(Deptid) AS total_departments,
->     GROUP_CONCAT(deptname) AS department_list
-> FROM department
-> GROUP BY Iname
-> ORDER BY Iname ASC;
+-----+-----+-----+
| institute_name | total_departments | department_list |
+-----+-----+-----+
| SCHC          | 1                | Health Sciences |
| SIBM          | 1                | Management     |
| SICSR         | 1                | Computer Applications |
| SIT           | 4                | CS,IT,Civil,E&TC |
| SLS           | 1                | Law            |
| SSBS          | 1                | Banking        |
| SSLA          | 1                | Liberal Arts   |
+-----+-----+-----+
7 rows in set (0.01 sec)
```

5. Give citywise seller details.

```
mysql> SELECT
->     city,
->     COUNT(Sid) AS number_of_sellers,
->     GROUP_CONCAT(slname) AS seller_names
-> FROM seller
-> GROUP BY city
-> ORDER BY city ASC;
+-----+-----+-----+
| city          | number_of_sellers | seller_names |
+-----+-----+-----+
| Bangalore     | 2                | College Book Store,pragati books store |
| Delhi         | 1                | Kavita      |
| Hyderabad    | 1                | Knowledge Hub |
| Mumbai       | 1                | Geeta       |
| Nagpur       | 1                | Academic House |
| Nashik       | 1                | Vidyarthi Book Depot |
| Noida        | 1                | Anita      |
| Pune         | 5                | Pragati Book Store,Pustak Seva,Mamta,Pioneer Book House,December Vendors |
+-----+-----+-----+
8 rows in set (0.00 sec)
```

6. Give author wise book details that have authored more than 2 books.

```
mysql> SELECT
->     a.Aname,
->     COUNT(w.Bid) AS total_books_authored
-> FROM author a
-> JOIN writes w ON a.Aid = w.Aid
-> GROUP BY a.Aname
-> HAVING COUNT(w.Bid) > 2
-> ORDER BY total_books_authored DESC;
```

Aname	total_books_authored
Yashavant Kanetkar	3

1 row in set (0.00 sec)

7. Give book details library wise whose price is less than 1000

```
mysql> SELECT
->     l.lname AS library_name,
->     COUNT(b.Bid) AS total_affordable_books,
->     GROUP_CONCAT(b.Bname) AS book_titles
-> FROM library l
-> JOIN books b ON l.Lid = b.Lid
-> WHERE b.Price < 1000
-> GROUP BY l.lname
-> ORDER BY l.lname ASC;
```

library_name	total_affordable_books	book_titles
SIBM Library	1	Management Principles
SICSR Library	1	Software Engineering
SIOM Library	1	Operating System Concepts
SIIT Library	11	Database Management Systems, Computer Networks, Software Engineering Basics, Applied Thermodynamics, Programming Language, Database System Concepts, Operating Systems, Let Us C, Understanding Pointers in C, Let Us C++, Database Management Systems
SLS Library	1	Constitutional Law of India
SSBS Library	1	Advanced Macroeconomics

6 rows in set (0.00 sec)

8. Give department wise staff details.

```
mysql> SELECT
->     d.deptname,
->     COUNT(s.Stid) AS staff_count,
->     GROUP_CONCAT(s.stname) AS staff_members
-> FROM department d
-> JOIN staff s ON d.Deptid = s.deptid
-> GROUP BY d.deptname
-> ORDER BY d.deptname ASC;
```

deptname	staff_count	staff_members
Banking	1	Meera Nair
Civil	1	Neha Sharma
Computer Applications	1	Priti Shah
CS	1	Amit Joshi
E&TC	1	Anjali Rao
Health Sciences	1	Asha Menon
IT	1	Sunil Patil
Law	1	Sanjay Gupta
Liberal Arts	1	Rakesh Singh
Management	1	Satish Kumar

10 rows in set (0.00 sec)

9. How many books are issued library wise

```
mysql> SELECT
->     l.lname AS library_name,
->     COUNT(i.issueid) AS total_books_issued
-> FROM ilibrary l
-> JOIN issue i ON l.Lid = i.Lid
-> GROUP BY l.lname
-> ORDER BY total_books_issued DESC;
```

library_name	total_books_issued
SIT Library	5
SLS Library	2
SIBM Library	1
SICSR Library	1
SSBS Library	1
SIOM Library	1
SCHC Library	1
SSLA Library	1
SIMS Library	1

9 rows in set (0.00 sec)

10. Give purchase details publisher wise.

```
mysql> SELECT
->     pub.Pname AS publisher_name,
->     COUNT(pr.prid) AS total_transactions,
->     SUM(pr.quantity) AS total_books_bought,
->     SUM(pr.totalcost) AS total_amount_spent
-> FROM publisher pub
-> JOIN purchase pr ON pub.Pid = pr.Pid
-> GROUP BY pub.Pname
-> ORDER BY total_amount_spent DESC;
```

publisher_name	total_transactions	total_books_bought	total_amount_spent
Cambridge University Press	1	5	67500.00
McGraw Hill	3	21	52600.00
Wiley Publications	1	4	50000.00
Springer	1	6	39000.00
Elsevier	1	9	16200.00
Technical Publications	2	25	16185.00
Oxford University Press	2	22	14950.00
Winter Books Publishing	1	20	10000.00
Pearson	1	5	3750.00

9 rows in set (0.00 sec)

11. Display books in a descending order of their cost.

```
mysql> SELECT
->      Bname,
->      Price
-> FROM books
-> ORDER BY Price DESC;
```

Bname	Price
Constitutional Law	13500.00
Advanced Artificial Intelligence	12500.00
Introduction to Algorithms	11000.00
Cloud Computing Essentials	6500.00
Business Analytics	1800.00
Data Structures Using C	1200.00
Database Management Systems	950.00
Operating Systems	950.00
Operating System Concepts	850.00
Database System Concepts	850.00
Software Engineering	799.00
Computer Networks	750.00
Constitutional Law of India	750.00
Advanced Macroeconomics	600.00
Database Management Systems	600.00
Software Engineering Basics	550.00
Management Principles	500.00
Let Us C++	500.00
The C Programming Language	450.00
Let Us C	450.00
Applied Thermodynamics	420.00
Understanding Pointers in C	350.00

22 rows in set (0.00 sec)

Perform a study assignment on SQL Views which should include following details:

12. Need of SQL Views

A View is a virtual table created by a query joining one or more tables. The primary needs for views are:

- **Security:** To hide sensitive columns (like an employee's salary) by only including non-sensitive columns in the view.
- **Simplicity:** To turn a highly complex query with multiple joins (like linking books, library, and writes) into a simple, single virtual table that users can easily query.
- **Reusability:** You write the complex SQL once and save it as a view, preventing the need to rewrite the same long query multiple times.

13. How to implement and use views in MYSQL.

implement a view using the CREATE VIEW statement.

Creation syntax:

CREATE VIEW view_name AS SELECT columns FROM tables WHERE conditions;

Usage syntax:

Once created, query it exactly like a normal table:

SELECT * FROM view_name;

14. Write a view to select all books of SIT library whose cost is less than 1000/-

```
mysql> CREATE VIEW SIT_Affordable_Books AS
-> SELECT
->     b.Bid,
->     b.Bname,
->     b.Price,
->     l.lname AS library_name
-> FROM books b
-> JOIN ilibrary l ON b.Lid = l.Lid
-> WHERE l.lname = 'SIT Library' AND b.Price < 1000;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from SIT_Affordable_Books;
+-----+-----+-----+-----+
| Bid | Bname                                | Price | library_name |
+-----+-----+-----+-----+
| 201 | Database Management Systems         | 950.00 | SIT Library  |
| 202 | Computer Networks                   | 750.00 | SIT Library  |
| 453 | Software Engineering Basics         | 550.00 | SIT Library  |
| 881 | Applied Thermodynamics              | 420.00 | SIT Library  |
| 999 | The C Programming Language          | 450.00 | SIT Library  |
| 1001 | Database System Concepts            | 850.00 | SIT Library  |
| 1002 | Operating Systems                   | 950.00 | SIT Library  |
| 1003 | Let Us C                            | 450.00 | SIT Library  |
| 1004 | Understanding Pointers in C         | 350.00 | SIT Library  |
| 1005 | Let Us C++                          | 500.00 | SIT Library  |
| 4444 | Database Management Systems         | 600.00 | SIT Library  |
+-----+-----+-----+-----+
11 rows in set (0.00 sec)
```

Conclusion: Through this assignment, I learned how to organize and filter database records using GROUP BY, HAVING, and ORDER BY. I also got to see how useful SQL Views are in real life. Basically, Views act like saved shortcuts for long, complicated queries. They make the database much easier to work with, help keep sensitive information secure, and let us reuse code without taking up extra storage space.